

## CANAVERAL EAST BUOY 41010, USA

WMO: 995460

Lat: 28.878N

Lon: 78.485W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 55.4       | 58.0 | 38.9                          | 34.9 | 57.4 | 40.9 | 37.7 | 59.0 | 35.9                  | 62.0 | 33.8 | 64.2 | 20.8                  | 300  | 0.530 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 3.6                    | 85.4            | 78.1 | 84.9 | 77.8 | 84.4 | 77.5 | 80.1                | 83.8 | 79.4 | 83.5 | 78.9 | 83.3 | 9.6                  | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 78.5                            | 148.1 | 83.3 | 78.1 | 145.8 | 83.1 | 77.4 | 142.6 | 82.8 | 43.3          | 84.0 | 42.7 | 83.5 | 42.2 | 83.4 | 84.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 32.0              | 29.1 | 24.8 | 51.3                       | 86.3 | 2.6                | 1.4 | 49.5                                               | 87.3 | 48.0       | 88.1 | 46.6       | 88.9 | 44.7       | 89.8 |
|                   |      |      | 45.0                       | 80.6 | 2.2                | 1.1 | 43.4                                               | 81.3 | 42.2       | 82.0 | 40.9       | 82.5 | 39.3       | 83.3 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 75.2   | 68.1  | 68.1  | 69.9  | 72.4  | 76.1  | 79.5  | 81.7  | 82.3  | 81.1  | 78.1  | 73.4  | 70.7 |
|                                                                               | DBStd       | 6.04   | 4.75  | 4.50  | 3.70  | 2.90  | 2.20  | 1.99  | 1.78  | 1.59  | 1.50  | 2.66  | 3.59  | 4.49 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 74     | 28    | 23    | 9     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 12   |
|                                                                               | CDD50       | 9181   | 560   | 507   | 617   | 671   | 810   | 885   | 982   | 1002  | 934   | 870   | 701   | 642  |
|                                                                               | CDD65       | 3779   | 123   | 110   | 161   | 221   | 345   | 435   | 517   | 537   | 484   | 405   | 253   | 188  |
|                                                                               | CDH74       | 29513  | 79    | 43    | 126   | 408   | 1883  | 4247  | 6014  | 6545  | 5462  | 3366  | 963   | 377  |
|                                                                               | CDH80       | 5971   | 0     | 0     | 0     | 0     | 14    | 561   | 1680  | 2151  | 1290  | 268   | 7     | 0    |
| Wind                                                                          | WSAvg       | 13.7   | 16.0  | 16.1  | 15.5  | 14.2  | 12.4  | 11.4  | 11.0  | 11.1  | 12.0  | 14.1  | 15.1  | 15.5 |
| Precipitation                                                                 | PrecAvg     | 43.4   | 3.1   | 2.7   | 3.0   | 2.6   | 2.6   | 4.7   | 3.8   | 4.9   | 5.8   | 4.6   | 2.9   | 2.8  |
|                                                                               | PrecMax     | 60.2   | 8.5   | 5.6   | 7.1   | 6.8   | 7.3   | 13.7  | 6.7   | 9.4   | 12.4  | 12.4  | 8.1   | 8.2  |
|                                                                               | PrecMin     | 26.1   | 0.9   | 0.4   | 0.4   | 0.8   | 0.3   | 1.2   | 0.8   | 1.0   | 1.3   | 0.9   | 0.7   | 0.7  |
|                                                                               | PrecStd     | 8.0    | 2.0   | 1.3   | 1.8   | 1.3   | 1.7   | 3.1   | 1.6   | 2.2   | 3.0   | 3.1   | 1.9   | 1.8  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 76.8  | 76.3  | 77.4  | 79.3  | 81.3  | 84.5  | 85.9  | 86.4  | 85.5  | 84.3  | 81.0  | 78.6 |
|                                                                               |             | MCWB   | 71.8  | 71.7  | 73.5  | 74.4  | 75.4  | 77.4  | 78.3  | 78.3  | 77.5  | 77.6  | 74.6  | 73.4 |
|                                                                               | 2%          | DB     | 75.9  | 75.3  | 76.3  | 78.1  | 80.4  | 84.1  | 85.0  | 85.8  | 84.5  | 83.1  | 80.1  | 78.0 |
|                                                                               |             | MCWB   | 71.0  | 70.9  | 72.3  | 72.8  | 74.9  | 77.4  | 78.0  | 78.2  | 77.1  | 77.0  | 74.2  | 72.9 |
|                                                                               | 5%          | DB     | 75.1  | 74.5  | 75.5  | 77.2  | 79.9  | 83.4  | 84.4  | 85.2  | 84.1  | 82.3  | 79.3  | 77.1 |
|                                                                               |             | MCWB   | 70.0  | 69.8  | 71.4  | 71.8  | 74.6  | 77.2  | 77.6  | 78.1  | 76.9  | 75.7  | 73.5  | 72.0 |
|                                                                               | 10%         | DB     | 74.1  | 73.5  | 74.7  | 76.3  | 79.3  | 82.6  | 84.2  | 84.7  | 83.5  | 81.5  | 78.2  | 76.2 |
|                                                                               |             | MCWB   | 68.7  | 68.5  | 70.0  | 70.9  | 74.1  | 76.7  | 77.5  | 77.8  | 76.6  | 74.6  | 71.7  | 70.7 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 73.1  | 73.5  | 75.4  | 75.6  | 77.8  | 80.2  | 80.4  | 81.2  | 79.9  | 79.0  | 77.4  | 75.2 |
|                                                                               |             | MCDB   | 76.2  | 75.2  | 76.5  | 79.0  | 80.1  | 82.7  | 84.4  | 84.3  | 83.1  | 83.1  | 79.6  | 77.8 |
|                                                                               | 2%          | WB     | 71.9  | 72.5  | 74.0  | 74.0  | 76.6  | 79.3  | 79.5  | 80.3  | 79.0  | 77.9  | 76.1  | 74.1 |
|                                                                               |             | MCDB   | 75.5  | 75.0  | 76.0  | 77.4  | 79.4  | 82.3  | 83.6  | 84.1  | 82.9  | 82.3  | 79.4  | 77.4 |
|                                                                               | 5%          | WB     | 70.7  | 71.3  | 72.6  | 72.9  | 75.7  | 78.5  | 79.1  | 79.5  | 78.4  | 77.1  | 74.8  | 73.2 |
|                                                                               |             | MCDB   | 74.6  | 74.5  | 75.4  | 76.4  | 79.0  | 82.1  | 83.4  | 84.0  | 82.8  | 81.6  | 79.1  | 77.0 |
|                                                                               | 10%         | WB     | 69.2  | 69.9  | 71.1  | 72.0  | 74.9  | 78.0  | 78.5  | 79.0  | 78.0  | 76.2  | 73.3  | 72.1 |
|                                                                               |             | MCDB   | 73.4  | 73.5  | 74.6  | 75.8  | 78.6  | 82.0  | 83.2  | 83.9  | 82.7  | 81.0  | 78.3  | 76.3 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 4.8   | 4.9   | 3.9   | 3.5   | 2.8   | 3.2   | 3.3   | 3.6   | 3.3   | 3.1   | 3.7   | 4.2  |
|                                                                               |             | MCDBR  | 5.2   | 5.0   | 4.9   | 4.3   | 3.2   | 3.2   | 3.0   | 3.5   | 3.7   | 3.7   | 4.1   | 4.5  |
|                                                                               |             | MCWBR  | 6.6   | 7.5   | 6.2   | 5.8   | 3.6   | 3.0   | 2.7   | 2.9   | 3.2   | 3.4   | 4.6   | 5.4  |
|                                                                               | 5% WB       | MCDBR  | 5.4   | 5.5   | 4.9   | 4.1   | 2.5   | 3.2   | 3.2   | 3.7   | 3.6   | 3.2   | 3.7   | 4.6  |
|                                                                               |             | MCWBR  | 7.3   | 7.7   | 6.7   | 5.8   | 3.8   | 3.6   | 3.3   | 3.5   | 3.4   | 3.6   | 5.2   | 5.8  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.361  | 0.372 | 0.380 | 0.402 | 0.436 | 0.449 | 0.461 | 0.452 | 0.421 | 0.401 | 0.367 | 0.360 |      |
|                                                                               | taud        | 2.523  | 2.507 | 2.476 | 2.381 | 2.334 | 2.374 | 2.356 | 2.392 | 2.453 | 2.482 | 2.533 | 2.556 |      |
|                                                                               | Ebn at Noon | 278    | 285   | 289   | 284   | 273   | 267   | 264   | 266   | 272   | 271   | 274   | 273   |      |
|                                                                               | Edh at Noon | 29     | 31    | 34    | 39    | 41    | 39    | 40    | 38    | 35    | 32    | 28    | 27    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 989    | 1262  | 1627  | 1974  | 2119  | 2036  | 1935  | 1864  | 1613  | 1402  | 1117  | 936   |      |
|                                                                               | RadStd      | 91     | 121   | 136   | 88    | 148   | 197   | 171   | 148   | 108   | 108   | 73    | 69    |      |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                       |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page